

JOONSUP KIM

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EDUCATION

Seoul National University, Graduate School of Data Science

M.S. in Data Science; Advisor: Professor Min-hwan Oh

Thesis / manuscript in progress: "Matching Bandits in Large Markets" (working title).

Seoul, South Korea

Expected February 2027

Yonsei University

B.S. in Mathematics; B.A. in Economics (Double Major)

Seoul, South Korea

February 2025

- **Mathematics Department Rank:** 4th of 42
- **Overall GPA:** 3.95/4.30 (converted: 3.85/4.00)
- **Major GPA:** 4.03/4.30 (converted: 3.92/4.00)
Mathematics: 3.94/4.30 (converted: 3.89/4.00) *Economics:* 4.14/4.30 (converted: 3.95/4.00)

RESEARCH INTERESTS

Operations management, revenue management, matching markets, bandit algorithms, online learning, reinforcement learning, and sequential decision-making under uncertainty.

RESEARCH EXPERIENCE

Seoul National University, Graduate School of Data Science

Research Assistant and M.S. Researcher; Advisor: Professor Min-hwan Oh

Seoul, South Korea

July 2024 – Present

- Developing a project on matching bandits in large markets, including problem formulation, algorithm design, theoretical analysis, and computational experiments.
- Studying bandit algorithms, online learning, reinforcement learning, and optimization for sequential decision-making under uncertainty.

Computational Science and Engineering Laboratory, Yonsei University

Undergraduate Research Assistant; Advisor: Professor Ji-hyun Lee

Seoul, South Korea

December 2021 – September 2022

- Analyzed theoretical epidemic models and studied how intervention and vaccination policies affect infection outcomes and economic costs.

RESEARCH IN PROGRESS

"Matching Bandits in Large Markets" (working title), with Professor Min-hwan Oh. M.S. thesis and manuscript in preparation; developing the market model, learning algorithm, theoretical analysis, and computational evaluation.

SELECTED PROJECTS AND COMPETITIONS

Kakao AI Agent Competition – 2nd Place (KRW 3,000,000)

December 2025

- Developed a persona-aware AutoReply Agent and a contextual-bandit Adaptive Push policy that learned push/no-push decisions from user reactions.

LG CNS Optimization Grand Challenge – Honorable Mention, Top 10% (KRW 500,000)

July 2025

- Modeled a multi-vehicle loading, unloading, and repositioning problem as a graph and combined the Hungarian algorithm with evolutionary and local-search methods.

TEACHING EXPERIENCE

Teaching Assistant, *Probability and Statistics for Data Science*, Seoul National University

July 2025

Teaching Assistant, *Machine Learning and Deep Learning I* (graduate course), Seoul National University

Fall 2025

Teaching Assistant, *Python for Data-Driven Business and Economics* (English-language course)

July 2026

AWARDS AND SCHOLARSHIPS

Academic Excellence Scholarship, Yonsei University (KRW 1,680,000)

2024

Yonsei Excellence Award, top 10% of college-year cohort

2024

Yonsei Excellence Award, top 10% of college-year cohort

2022

Academic Excellence Scholarship, Yonsei University (KRW 2,000,000)

2022

Mathematics Alumni Scholarship, Yonsei University (KRW 2,500,000)

2021

Yonsei Honor Award, top 3% of college-year cohort

2019

ADDITIONAL INFORMATION

Test Scores: GMAT 735; TOEFL 5.5 **Programming:** Python, C, C++, SQL

Languages: Korean (native); English (fluent; lived in Davis, California, for six years)

LEADERSHIP AND SERVICE

Vice President, Freesoul Surfing Club

July 2022 – February 2023

Academic Tutor in mathematics and science for underserved middle and high school students

March 2021 – May 2021